Multi-Axis Failure Modes in Simulator Transfer Diagnostics from Discrete Maneuver Tokens and Grounded Selection

Alba Maria Tellez Fernandez
amtellezfernandez@gmail.com
Waymo Open Dataset E2E Challenge

Abstract: We study a transfer failure mode in closed-loop imitation learning: a policy can appear well calibrated inside its training simulator while selecting geometrically incompatible actions after transfer through a different observation and dynamics interface. The controlled setting is a discrete autonomous-driving token library with a geometry-driven expert, Spotlight Reflex. On a 1080-rollout held-out Latin-hypercube generator sweep, plain continuous, token, and recurrent BC reach only 77–78% pass rate with 3.0–4.3% Proximity Violations, while two iterations of DAgger-style relabeling nearly close the internal frontier at 94.54% pass and 0.19% PV. This internal success does not reduce to one transfer behavior. In a controlled proxy-perturbation sweep (288 matched scenario-perturbation cases, 1440 agent rollouts), grounded vetoing strongly repairs three-step actor latency (collision 0.396 to 0.146; lane violation 0.875 to 0.208), but under route-offset corruption it improves lane adherence while degrading pass/collision. Waymo AlpaSim validation shows the same multi-axis structure externally. In the initial 10-clip matrix, clamping repairs offroad/wrong-lane behavior while increasing collision, hard vetoing repairs progress/offroad while vetoing 1987/1990 DAgger argmax decisions, and source decay helps wrong-lane behavior without repairing offroad or collision. Actor completion is not yet a positive method: the apparent 0.700 to 0.400 collision reduction is a score-cutoff world-frame oracle diagnostic, weakens to 0.700 to 0.600 under raw full-rollout oracle evaluation, and the latest same-pass raw rear-risk actor-axis method leaves collision unchanged at 0.700 while increasing progress from 0.216 to 0.471 and worsening wrong-lane from 0.300 to 0.400. These non-co-monotone interventions show that simulator transfer error decomposes into separate axes of geometry, safety, progress, and lane adherence rather than a single scalar robustness deficit. This identifies a narrower grounding issue: making the state representation more physical is not enough if the action-ranking operator remains miscalibrated under transfer. Frozen world/VLA priors on the Waymo E2E candidate-selection track isolate the placement of grounding: scalar-only selection reaches 7.695 RFS, generic latent fusion remains weak, a nonlinear head over Cosmos 64d embeddings reaches 7.845 RFS, and adding a latent world-prior plus direct selector reaches 7.848 RFS. The combined evidence supports a grounded-selection thesis: internal closed-loop success and offline accuracy are weak transfer proxies unless world priors are tied to per-axis candidate ranking and geometric disagreement is instrumented directly.

Keywords: autonomous driving, imitation learning, DAgger, simulator transfer, discrete action space, calibration, ManeuverToken, geometric world-state

10th Conference on Robot Learning (CoRL 2026), Austin TX, USA.

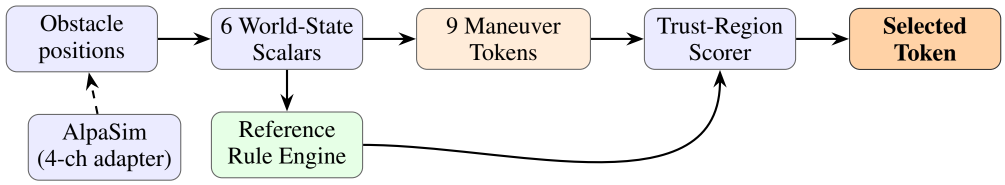

Figure 1: Spotlight Reflex pipeline. Obstacle positions (raw geometry, or via a 4-channel AlpaSim adapter) feed six scalar encoders. Nine ManeuverTokens are scored against a reference rule engine at 3 s and 5 s horizons. The highest-scoring token is executed. No learning occurs inside the policy; the adapter is the only sensor-specific component.

1 Introduction

Long-tail failure is a central open problem in autonomous driving. Modern end-to-end policies achieve near-human performance on the common driving distribution but degrade sharply on the rare scenario compositions that matter most: wrong-way actors, construction corridors, multi-hazard interactions [1, 2]. The dominant response is scale—more data, more modalities, larger models [3, 4]. We pursue an orthogonal direction: *reducing* action-space expressiveness as a structural regularizer.

The argument is topological rather than architectural. In a multi-hazard scene, the safe trajectory set can split into disconnected components: for example, pass left, pass right, or stop before the conflict. A continuous regressor moving between components can produce an intermediate trajectory that belongs to neither component and intersects the unsafe set. A discrete selector avoids this interpolation path by choosing among maneuver representatives with explicit clearance margins. Policy uncertainty changes which representative is chosen; it does not create a new trajectory between representatives.

This shifts the key question for grounded autonomy. Recent latent VLA systems correctly argue that driving should reason in geometry and dynamics rather than in text. The missing interface is the last step: how a grounded state representation is converted into a concrete action under proxy-state error. Our experiments show that this step cannot be audited by offline accuracy or one aggregate planning score. The same intervention can improve route geometry while worsening collision, or improve progress while leaving wrong-lane behavior unchanged. The transfer-critical object is therefore the *ordering* induced over bounded candidates, measured separately along safety, route, lane, and progress axes.

We make six contributions (Fig. 1):

1. **Non-interpolation theorem (Sec. 3.2).** We state a conditional safety result: if tokens lie inside disconnected safe maneuver classes with clearance margin greater than execution noise, discrete selection cannot enter the unsafe set by interpolation, while continuous transitions between classes must cross it.
2. **Geometric World-State Encoding (Sec. 3.4).** Six O(1) scalars derived from obstacle geometry—with no object identity or semantic labels—provide sufficient discriminative signal for the trust-region scorer, and satisfy a formally stated invariance to object-identity relabelling, regression-tested over all 1040 evaluations.
3. **Held-out generator evidence with strong learned baselines (Sec. 4.2).** Fixed-suite evaluations show that plain BC fails even with the same tokens and state features. A held-out Latin-hypercube generator sweep then shows the sharper frontier: two-step DAgger matches Spotlight Reflex on pass rate and improves PV rate, while one-step DAgger is weaker, three-step DAgger regresses, and trajectory scorers over token candidates do not close the stress-suite gap.
4. **Multi-axis transfer failure in AlpaSim (Sec. 4.3).** A 10-clip external transfer matrix shows that clamping, source decay, and hard geometric vetoes repair different axes and damage

others. A matched world-frame actor-oracle rerun localizes part of the collision surface to missing dynamic actors, while an explicit actor-axis proxy remains inconclusive as a positive method. Transfer failure is therefore not a single scalar degradation: offroad, wrong-lane, collision, progress, and distance-to-ground-truth expose distinct mechanisms.

5. **Controlled proxy-perturbation diagnostic (Sec. 4.4).** Inside the internal simulator, we perturb only the planner-visible proxy state while keeping dynamics fixed. A 288-case matched sweep (1440 agent rollouts) shows that latency, heading-bias, route-offset, lane-scale, and feature-noise corruption induce different intervention effects, reducing the concern that the AlpaSim result is purely an adapter artifact.
6. **Grounded selection evidence from frozen visual/world priors.** In the Waymo E2E candidate-selection track, scalar-only selectors plateau at 7.695 RFS, InternVLA-only grounding does not help, and linear fusion of InternVLA with Cosmos remains weak. A nonlinear direct-policy head over frozen Cosmos 64d embeddings reaches 7.845 RFS, while a latent world-prior plus direct selector reaches 7.848 RFS with lower regret to oracle. This turns the VLA comparison into a placement result: frozen geometry/world priors help when they participate in the selector that ranks bounded candidates, but not when they are appended as generic latent features.

The strength of these claims comes from the controls, not from aggregate score chasing. Across the learned-policy variants, the token library, low-level controller, and AlpaSim scene set are held fixed; clamping, vetoing, source decay, and axis constraints change only the selection interface. In the proxy-perturbation sweep, simulator dynamics and collision evaluation are held fixed while only the planner-visible reconstructed state is corrupted. In the WOD grounding ablation, the candidate pool and validation frames are held fixed while the latent prior and selector head vary. These controls let us localize failure to action ranking under reconstructed geometry, rather than to token availability, vehicle control, or a single simulator idiosyncrasy.

2 Related Work

Long-tail failure in imitation learning. Pomerleau [5] and Ross et al. [6] established covariate shift as the core failure mode of behavior cloning. Ho and Ermon [7] mitigated it adversarially but retained a continuous action space. Bansal et al. [8] synthesised worst-case training scenarios; Dauner et al. [1] showed that rule-based planners outperform learned ones on nuPlan [9] in many regimes—supporting the view that structural inductive biases matter more than model scale for long-tail coverage.

Discrete action spaces. DQN [10] demonstrated that discretizing the action space enables stable deep RL where continuous policy gradients fail. Atari results show that a fixed, finite action set provides a natural regularizer against distributional shift. We apply this structural idea to the planning setting without RL: ManeuverTokens provide a finite, semantically grounded action set whose footprints are invariant to scene uncertainty.

Potential-field and reactive planners. Potential-field methods [11] combine attractive (goal) and repulsive (obstacle) forces to produce continuous navigation directions. Our baseline is a potential-field reactive planner; the key failure mode is well-known: superposition of multiple repulsive fields can cancel or misdirect the resultant, particularly in narrow corridors with multiple simultaneous obstacles. ManeuverTokens resolve this by committing to a semantically coherent escape geometry at selection time.

Geometric abstraction for planning. Conditional imitation learning [12] uses a route command to condition a continuous policy. Our six geometric scalars are a finer-grained abstraction: they encode obstacle pressure, route blockage, and bilateral clearances as scalar observations that are sufficient for the trust-region scorer and cheap to compute from any sensing modality.

Table 1: ManeuverToken definitions. Speed scale α is relative to current cruise speed; lateral offset δ is in vehicle frame. *Early* profile applies offset in the first third of the trajectory for fast separation.

Token	α	δ (m)	Profile	Semantic
STOP	0.00	0.0	—	Emergency stop
CRAWL	0.15	0.0	smooth	Cautious forward
MAINTAIN	1.00	0.0	smooth	Current speed, lane centre
SLOW_YIELD	0.50	0.0	smooth	Yield, hold position
NUDGE_LEFT	0.80	+0.9	smooth	Mild lateral evasion left
NUDGE_RIGHT	0.80	−0.9	smooth	Mild lateral evasion right
EVASIVE_LEFT	0.70	+2.2	early	Strong evasion left
EVASIVE_RIGHT	0.70	−2.2	early	Strong evasion right
LANE_RECOVER	0.90	0.0	smooth	Return to lane centre

VLA grounding and latent world priors. Recent autonomous-driving VLA work argues that direct trajectory generation and textual chain-of-thought are not enough: latent reasoning must be grounded in geometric and dynamic priors. LaST-VLA [13], for example, distills 3D geometric features and video-world-model dynamics into latent tokens and optimizes trajectory-level metrics with RL. Our study is complementary rather than competitive. We do not train a large VLA; instead, we test where such priors should enter a transfer-critical driving stack. The missing variable is not just representation quality; it is placement. In our WOD ablation, InternVLA-only features and linear Cosmos–InternVLA fusion remain below the scalar baseline, while a selector-aligned nonlinear Cosmos head and a latent world-prior plus direct selector produce the best tracked RFS and regret. The link to LaST-VLA is therefore operational: physical latent grounding matters most when it constrains the action-ranking operator over bounded candidates, the same stage that fails under AlpaSim proxy-state transfer. Without that placement test, aggregate planning scores can hide non-co-monotone failures across collision, route geometry, lane adherence, and progress.

Simulator transfer and validation scope. Khurana et al. [14] and Shi et al. [15] study trajectory prediction in sensor-realistic environments. Our AlpaSim adapter is a transfer diagnostic that decouples policy logic from sensor processing; it is not a substitute for physics simulation or real-vehicle validation.

3 Method

3.1 Discrete ManeuverToken Space

We define nine ManeuverTokens (Table 1) covering the evaluated safety-critical response classes of the SE(2) maneuver manifold. Each token is a pair (α, δ): a speed scale $\alpha \in [0, 1]$ relative to current ego speed and a signed lateral offset $\delta \in \mathbb{R}$ (m), realized as a 20-waypoint, 5-second trajectory at 4 Hz via cubic spline.

Coverage for the evaluated hazard family. The nine tokens are not claimed to be globally optimal. They are a minimal engineering instantiation for the hazard family evaluated here: hard stop, cautious forward progress, nominal progress, yield-without-stopping, bilateral mild avoidance, bilateral strong avoidance, and post-avoidance recovery.

Proposition 1 (Token removal induces a maneuver gap)**.** *For the evaluated multi-hazard family, removing any one response class from the 9-token library eliminates at least one feasible safe response mode: longitudinal hard stop, cautious progress, mild lateral avoidance, strong lateral avoidance, or recovery. The resulting controller must either overreact (e.g., full evasion for a mild obstacle), underreact (e.g., nudge where full evasion is required), or fail to re-enter the lane.*

Proposition 2 (Geometry-Invariance)**.** *Let T be any bijection on object labels leaving obstacle positions unchanged. Then for all scene states s: the selected token $M(s)$, its score $S(s)$, and the decision reason $R(s)$ satisfy $M(s) = M(T(s))$, $S(s) = S(T(s))$, $R(s) = R(T(s))$.*

The geometry-invariance proposition follows from the encoding in Sec. 3.4: no token score depends on object labels. We regression-test it over all 1040 evaluations by relabelling all obstacles in every scene and asserting equality of outputs.

3.2 Topological Safety Through Non-Interpolation

Let $\mathcal{A}$ denote the action space of 5 s trajectories, and for scene state s let $\mathcal{U}(s) \subset \mathcal{A}$ be the unsafe set (trajectories violating the clearance constraint). Let $\mathcal{S}(s) = \mathcal{A} \setminus \mathcal{U}(s)$ be the safe set. In multi-hazard scenes, $\mathcal{S}(s)$ may decompose into disjoint maneuver classes $\mathcal{S}_1(s), \ldots, \mathcal{S}_K(s)$, such as stop-before-conflict, pass-left, and pass-right.

We use the trajectory metric $d(\tau_a, \tau_b) = \max_t \|\tau_a(t) - \tau_b(t)\|_2$ over the 5 s horizon. A continuous policy outputs $a_c(s) \in \mathcal{A}$ and is executed with bounded decision or actuation error ϵ, $\|\epsilon\| \leq \eta$. A token policy selects $\tau_i \in T = \{\tau_1, \ldots, \tau_N\}$ and executes $\tilde{\tau}_i = \tau_i + \epsilon$.

Theorem 1 (Safety through non-interpolation)**.** *Fix a scene state s. Suppose:*

1. *each selected token τ_i lies in a safe maneuver class $\mathcal{S}_k(s)$;*
2. *each selected token has clearance margin $\mathrm{dist}(\tau_i, \mathcal{U}(s)) > \eta$;*
3. *there exist two safe maneuver classes $\mathcal{S}_i(s)$ and $\mathcal{S}_j(s)$ such that every continuous path in $\mathcal{A}$ from $\mathcal{S}_i(s)$ to $\mathcal{S}_j(s)$ intersects $\mathcal{U}(s)$.*

Then every noisy execution of a selected token remains safe, while any continuous controller that changes between $\mathcal{S}_i(s)$ and $\mathcal{S}_j(s)$ by interpolation must pass through an unsafe trajectory.

Proof. For a token, $\mathrm{dist}(\tau_i, \mathcal{U}(s)) > \eta$ implies that every perturbation with norm at most η remains outside $\mathcal{U}(s)$. For the continuous case, let $\gamma : [0, 1] \to \mathcal{A}$ be any continuous transition from a trajectory in $\mathcal{S}_i(s)$ to one in $\mathcal{S}_j(s)$. By assumption, γ intersects $\mathcal{U}(s)$, so there exists $t^\star$ with $\gamma(t^\star) \in \mathcal{U}(s)$. Thus interpolation between disconnected safe maneuver classes necessarily traverses the unsafe set, whereas token selection jumps between representatives and does not execute the intermediate path. □

The theorem is conditional: it does not prove that nine tokens are globally optimal, nor that the specific numeric offsets in Table 1 are unique. It identifies the property we test empirically: safety comes from selecting margin-separated maneuver classes, not from continuous interpolation through the action manifold.

3.3 Metric-Ordering Limits Under Transfer

The external diagnostic requires a second formal distinction. Let $z = \phi(x)$ be the internal simulator state representation and $z' = \phi'(x)$ be a transferred proxy representation of the same underlying driving scene. For a candidate token τ, define axis-specific costs $c_j(z', \tau)$ for collision risk, offroad risk, wrong-lane risk, progress loss, and tracking error, with lower values preferred. A selector intervention is monotone across axes only if replacing token τ_a with token τ_b weakly decreases all costs c_j under z'.

Proposition 3 (Non-co-monotone transfer axes)**.** *If two transfer metrics induce conflicting orderings over the token library, i.e. there exist tokens τ_a, τ_b and axes p, q such that*

$$c_p(z', \tau_a) < c_p(z', \tau_b) \quad \textit{and} \quad c_q(z', \tau_a) > c_q(z', \tau_b),$$

then no intervention that switches from τ_b to τ_a can be interpreted as a uniform transfer improvement. It improves axis p while worsening axis q.

Proof. The first inequality states that τ_a is better than τ_b on axis p; the second states that τ_a is worse than τ_b on axis q. Therefore the switch has opposite sign on the two metrics. A scalar rollout score that reports only the improvement on p, or only an aggregate of p and q, can hide this disagreement. □

Proposition 3 is deliberately modest. It does not claim that transfer failure is unsolvable. It states the condition under which a single intervention cannot be judged by one scalar success rate. The AlpaSim matrix in Table 6 is an empirical instance: clamping, hard vetoes, and source-decayed DAgger each improve at least one transfer axis while worsening or failing to move another.

3.4 Geometric World-State Encoding

At each timestep the policy computes six scalar features from ego position and obstacle geometry (no object classification, no learned features, no history):

$$\begin{aligned}
\textit{pressure} &= \mathrm{clip}\left(\frac{10 - d_{\min}}{10}, 0, 1\right) && (1)\\
\textit{blockage} &\in [0,1], \quad \text{fraction of forward corridor cross-section occupied} && (2)\\
\textit{blocked} &\in \{0,1\}, \quad \text{hard stop within braking distance} && (3)\\
\textit{left_clr} &= \text{free lateral space left at half vehicle width (m)} && (4)\\
\textit{right_clr} &= \text{free lateral space right at half vehicle width (m)} && (5)\\
\textit{escape} &= \arg\max(\textit{left_clr},\ \textit{right_clr}) && (6)
\end{aligned}$$

All six scalars are $O(1)$ given a sorted obstacle list ($d_{\min}$ by distance). The encoding contains no object identity or semantic labels, so geometry-invariance follows directly. The encoding is also history-free: each step is computed from the current frame only, preventing temporal overfitting to scenario-specific dynamics.

3.5 Trust-Region Scoring

Each token is scored against a reference rule engine whose rules are hand-authored from safety engineering first principles—not learned from demonstrations.

Speed-scaled tolerance.

$$s(v) = \mathrm{clip}\left(\frac{v - 1.4}{11.0 - 1.4},\ 0,\ 1\right) \times 0.5 + 0.5 \tag{7}$$

$s(v) \in [0.5, 1.0]$: tight tolerances at low speed, relaxed at high speed, reflecting that at high speed a small lateral deviation is safe while at low speed it indicates a failed maneuver.

Trust-region thresholds. At 3 s horizon: $\pm 1.0 \cdot s(v)$ m lateral, $\pm 4.0 \cdot s(v)$ m longitudinal. At 5 s: $\pm 1.8 \cdot s(v)$ m, $\pm 7.2 \cdot s(v)$ m.

Token score. Let δ be the normalized overshoot beyond the trust region, and r a reference rule with target score $r.\texttt{score} \in [0, 100]$:

$$\mathrm{score}_t(\tau, r) = \begin{cases} r.\texttt{score} & \delta = 0 \\ \max(r.\texttt{score} \cdot 0.1^{\delta},\ 4.0) & \delta > 0 \end{cases} \tag{8}$$

$$S(\tau) = 0.5 \cdot \max_r \mathrm{score}_{3\mathrm{s}}(\tau, r) + 0.5 \cdot \max_r \mathrm{score}_{5\mathrm{s}}(\tau, r) + P(\tau) \tag{9}$$

where $P(\tau)$ adds clearance penalties (-1000 if clearance < 0.55 m) and progress bonuses (+40 max). The token $\tau^* = \arg\max_\tau S(\tau)$ is executed.

Reference rules. Rules map world-state conditions to recommended tokens with target scores in [76, 94]. Key rules: *pressure* < 0.25 and open corridor → MAINTAIN (92); *pressure* > 0.20 with preferred escape → $\text{NUDGE}_{\text{escape}}$ (91); *pressure* ≥ 0.35 with preferred escape → $\text{EVASIVE}_{\text{escape}}$ (94); *blocked* $= 1$ → STOP (82). Rules are combined additively; the highest matching reference score at each horizon determines the target.

3.6 AlpaSim Proxy-State Adapter

Waymo's AlpaSim provides sensor-realistic front-camera input, route command, and ego dynamics. We bridge these observations to the same geometric scalar interface via four channels:

1. **Route → lane curvature.** LEFT/STRAIGHT/RIGHT generates a 5-point sigmoid spline (± 2.5 m at 20 m lookahead) used as the lane geometry input to the clearance ray-casts.
2. **Brightness → visibility augmentation.** vis $= \max(0, (0.35 - \mu_I)/0.35)$, where μ_I is mean frame brightness. Below 35% brightness, *pressure* is augmented proportionally.
3. **Ego dynamics → risk scalars.** brake $= \max(0, -a/4.0)$; low_speed $= \max(0, (1.5 - v)/1.5)$. These augment the reference rule thresholds.
4. **Structured hazards → obstacle primitives.** Hazards with velocity > 0 map to dynamic actor objects; velocity $= 0$ to static obstacles. Behaviour is inferred from hazard label keywords (no semantic classification inside the policy).

4 Experiments

4.1 Simulation Benchmark

Setup. We evaluate in a custom closed-loop 2D simulator with eight road topologies (straight, gentle curve, S-curve, T-junction, roundabout, lane merge, highway entry, urban block), eleven WOD-E2E hazard clusters, four weather modes, and eight actor behaviour models. The compositional OOD generator samples topology × hazard × weather independently: $P(\text{topo}, \text{hazard}, \text{weather}) = P(\text{topo}) \times P(\text{hazard}) \times P(\text{weather})$, yielding 352 structurally distinct compositions. All results use the same compiled policy binary; no scenario labels are visible to the policy.

Baseline. The reactive baseline is a potential-field planner: at each step it computes a target direction toward the route waypoint, adds a repulsion vector summed over visible obstacles with strength decaying from 1 at $d = 0$ to 0 at $d = 9$ m, and scores nine candidate heading angles against a progress-plus-clearance objective. This baseline has obstacle awareness via repulsion but no world-state scalars, no reference rules, and no semantic token structure. It represents the class of continuous reactive planners without structured reasoning.

Safety metric definition. We define a **Proximity Violation** as any evaluation in which the rollout does not reach the goal *and* at least one step has $C > 0.90$, where $C = \max(0, \min(1, (5 - d_{\text{min}})/5))$ and d_{min} is the minimum signed clearance (metres) between the ego-vehicle centre and any visible obstacle surface. $C > 0.90$ corresponds to $d_{\text{min}} < 0.25$ m. The metric is proximity-based rather than fault-based: it does not distinguish whether the ego vehicle moved into an obstacle or an obstacle moved into the ego vehicle. However, the simulator's actor behaviour models (crossing, cut-in, head-on approach, sudden braking) do not include rear-approach scenarios; every proximity violation therefore represents a failure of the ego policy to maintain safe separation, not an unavoidable passive impact. Rollouts that *reach the goal* with momentarily high C (close pass) are not flagged; only failed rollouts with proximity violations count.

Fixed-suite diagnostic. Table 2 shows Spotlight Reflex has 0 observed Proximity Violations (PVs) across all 1040 evaluations, corresponding to a Wilson 95% upper bound of 0.37%. This includes the 5.8% of evaluations where the policy stalled and did not reach the goal: even in those failed rollouts, clearance never dropped below 0.25 m. BC agents record PV rates of 3.8–4.4% on Gauntlet and 3.1–3.8% on Adversarial.

Table 3 shows the three-agent comparison with Wilson 95% CI. The Spotlight Reflex PV CI [0.0, 0.8%] on Gauntlet is statistically disjoint from both BC agents ([2.4, 5.8%] and [2.9, 6.6%]). Critically, **Continuous-BC and Token-BC produce near-identical PV rates**: providing the 9-token vocabulary

Table 2: Simulation results across four suites, seeds 1–80 (matched). Pass = scenario completion. "PV" = Proximity Violation ($C > 0.90$, $d < 0.25$ m, failed rollout only). Spotlight Reflex has 0 observed PVs across all 1040 evaluations (Wilson 95% upper bound 0.37%), including stall failures where the goal was not reached. BC agents record PVs on Gauntlet and Adversarial.

Suite	Runs	Pass (SR)	PV rate (SR)	PV rate (BC)
Compositional OOD (5 cases)	400	92.2%	0%	1.0–1.2%
Adversarial (2 cases)	160	91.9%	0%	3.1–3.8%
Gauntlet (6 cases, 4-hazard)	480	94.2%	0%	3.8–4.4%
Hidden holdout	80	91.2%	0%	—
Total (SR)	**1040**	**93.3%**	**0 PVs**	

Table 3: Three-agent comparison on matched Gauntlet and Adversarial stress suites, seeds 1–80. All agents use the identical 10-feature geometric input. BC agents trained on 143 k expert transitions with inverse-frequency class weights. "PV" = Proximity Violation ($C > 0.90$, $d < 0.25$ m, failed rollout only); Wilson 95% CI shown. Spotlight Reflex has 0 observed PVs in both suites—including stall failures where the goal was not reached but safe separation was maintained throughout.

	Gauntlet (n=480)		Adversarial (n=160)	
Agent	Pass	PV rate [95% CI]	Pass	PV rate [95% CI]
Continuous-BC (A)	77.7%	4.4% [2.9, 6.6]	76.2%	3.8% [1.7, 7.9]
Token-BC (B)	77.7%	3.8% [2.4, 5.8]	76.9%	3.1% [1.3, 7.1]
Spotlight Reflex (C)	**94.2%**	**0.0% [0.0, 0.8]**	**91.9%**	**0.0% [0.0, 2.3]**

to a learned policy does not prevent proximity violations. Only the geometric reference rules—enforcing that pressure ≥ 0.35 triggers EVASIVE over NUDGE—keep the observed PV count at zero. This supports the non-interpolation mechanism in Theorem 1: gradient descent on the same features approximates the escape rule, but does not enforce the margin-separated maneuver class.

4.2 Held-Out Latin-Hypercube Sweep

The fixed suites above are useful diagnostics, but they are hand-authored stress tests. For the primary OOD result we sample 12 held-out generator profiles with Latin-hypercube coverage over nine scenario axes: lane-width range, gauntlet pressure, adversarial pressure, hidden-suite pressure, visibility, actor intrusion scale, ambient obstacle count, and hidden hazard count. Each profile is evaluated on matched seeds 1–10 across Gauntlet, Adversarial, and Hidden suites, for 1080 closed-loop rollouts per agent.

We compare five agents using the same 10-feature geometric state: Continuous-BC, Token-BC, Token-RNN-BC, Token-DAgger-BC, and Spotlight Reflex. Token-DAgger-BC uses two rounds of on-policy relabeling: first rolling out Token-BC and relabeling visited states with Spotlight, then rolling out the resulting DAgger policy and relabeling again.

Table 4 changes the interpretation of the fixed-suite result. Plain BC and recurrent BC still fail under held-out generator shift, but two-step DAgger closes the gap to Spotlight Reflex: the paired overall pass delta is +0.09 percentage points and the PV delta is -0.46 percentage points relative to Spotlight. The learned policy is not uniformly better—Spotlight still has the best Gauntlet pass rate—but the frontier is no longer "rules beat learning." The correct conclusion is that on-policy correction can learn most of the geometric rule, while plain expert-state imitation cannot.

Table 5 shows the DAgger frontier is not monotonic. Iteration 2 improves the adversarial suite from 87.50% pass / 0.42% PV to 90.83% pass / 0.00% PV, but iteration 3 falls back to 87.92% pass / 0.42% PV and introduces hidden-suite PVs. This suggests that additional aggregation changes the selector in a harmful way, but not yet *why*. We therefore run targeted counter-ablations to distinguish token-frequency bias from late-pool aggregation pressure.

(a) Spotlight Reflex: wrong-way actor, night. Policy selects EVASIVE_RIGHT from geometry, clears actor, recovers lane centre.

(b) Construction zone: 5.2 m corridor, cone field, lane closure. Policy selects NUDGE_RIGHT. No training data for this composition.

Figure 2: Qualitative rollout key frames. Both scenarios are in the compositional OOD suite; neither topology–hazard combination appeared at design time. Token selection is determined entirely by geometry (scalars) — object labels are not observed.

Table 4: Held-out Latin-hypercube generator sweep, 12 profiles $\times$ matched seeds 1–10, Gauntlet/Adversarial/Hidden suites ($n = 1080$ per agent). Entries are pass/PV percentages. Two-step Token-DAgger-BC is the strongest learned baseline.

Agent	Overall	Gauntlet	Adversarial	Hidden
Continuous-BC	77.69 / 3.52	77.50 / 4.31	75.42 / 2.08	83.33 / 1.67
Token-BC	76.85 / 4.26	77.64 / 4.86	71.67 / 2.92	82.50 / 3.33
Token-RNN-BC	78.43 / 2.96	79.17 / 2.92	74.58 / 2.50	81.67 / 4.17
Token-DAgger-BC (2-step)	**94.54 / 0.19**	96.25 / 0.28	**90.83 / 0.00**	**91.67 / 0.00**
Spotlight Reflex	94.44 / 0.65	**96.67 / 0.28**	90.00 / 0.83	90.00 / 2.50

4.3 AlpaSim External Diagnostic

We use AlpaSim as an external diagnostic, not as a statistically powered safety benchmark. The purpose is to ask whether the internally strong learned selector remains calibrated when its 10-dimensional simulator state is reconstructed from a front-camera, route command, ego dynamics, and sparse hazard primitives. We evaluate matched 10-clip WOD-E2E matrices for which all compared policies complete the same AlpaSim launch path. Unless otherwise stated, AlpaSim numbers below are raw summaries from `aggregate/metrics_unprocessed.parquet`. We keep score-cutoff summaries in the release artifact as a separate diagnostic view, but do not use them as the primary method claim because they can hide incidents after AlpaSim has already applied collision/offroad or distance-to-ground-truth cutoffs.

Table 6 falsifies the stronger methods claim that one inference-time patch can convert the internally strong DAgger policy into a robust external policy. The unmodified two-step DAgger run fails on all major axes: 0.600 collision rate, 0.900 offroad rate, 0.700 wrong-lane rate, and only 0.034 mean progress. Lateral clamping directly repairs geometry, reducing offroad by 0.400 and wrong-lane by 0.500 while improving distance-to-ground-truth from 13.86 m to 5.21 m. But the same clamp raises collision rate from 0.600 to 0.700, so it is not a uniform transfer improvement. Adding axis constraints to the clamped selector pushes the geometry repair further: offroad falls to 0.200 and distance-to-ground-truth falls to 2.25 m. Collision still remains 0.700, which isolates the central external mechanism: route geometry can be repaired without repairing interaction safety.

The hard-veto hybrid repairs a different axis. It improves mean progress from 0.034 to 0.837 and offroad rate from 0.900 to 0.200, but raises collision rate to 0.800, leaves wrong-lane rate unchanged

Table 5: DAgger iteration study on the same held-out LHS sweep. Iteration 2 is best; iteration 3 regresses despite adding more relabeled states.

Policy	Overall	Gauntlet	Adversarial	Hidden
Token-DAgger (1-step)	93.98 / 0.46	96.81 / 0.56	87.50 / 0.42	90.00 / 0.00
Token-DAgger (2-step)	**94.54 / 0.19**	96.25 / 0.28	**90.83 / 0.00**	**91.67 / 0.00**
Token-DAgger (3-step)	92.50 / 0.46	94.31 / 0.28	87.92 / 0.42	90.83 / 1.67
Spotlight Reflex	94.44 / 0.65	96.67 / 0.28	90.00 / 0.83	90.00 / 2.50

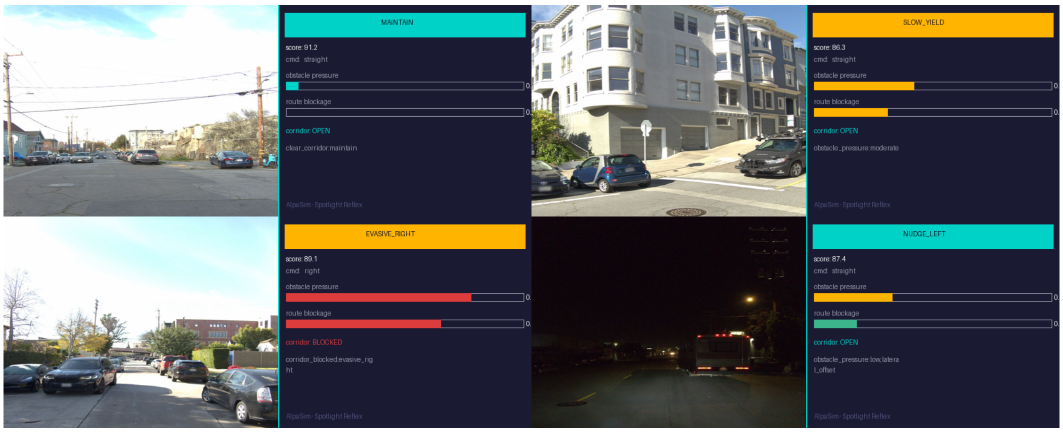

Figure 3: AlpaSim: real Waymo WOD-E2E front-camera frames alongside the 4-channel adapter's output. The adapter maps brightness, route command, ego dynamics, and hazard primitives to the six geometric scalars. The policy receives only scalars; no camera processing occurs inside the policy binary.

at 0.700, and worsens distance-to-ground-truth to 22.71 m. Its selection log shows why this is not evidence for a successful hybrid policy: the veto fires on 1987/1990 frames (99.85%, Wilson 95% CI [99.56, 99.95]), with 1827 vetoes due to geometric score gap and 160 due to geometric rank. Spotlight-style selection wins 1965 frames, DAgger wins only 2, and there are 16 compromise selections. The hard veto therefore recovers progress largely by collapsing back to the geometric scorer, not by combining two well-aligned policies.

The same logs also explain why stronger selector thresholds are the wrong remedy for AlpaSim safety. The structured-hazard interface is present in the adapter, but in this 10-clip front-camera transfer matrix the axis-constrained and hard-veto runs both record 0/1990 frames with structured hazards or moving actors. The scorer's top candidate is therefore interpreted as a clear corridor on 87.0% of axis-constrained frames and 90.7% of hard-veto frames. This localizes part of the collision failure upstream of token ranking: route geometry can be clamped or re-ranked, but interaction safety cannot be fully recovered from candidates scored against an actor-incomplete proxy state.

The world-frame actor-proxy probe fixes the earlier stale-frame oracle confound by storing actors in world coordinates and transforming them into the current rollout ego frame at inference time. Table 8 separates its two diagnostic metric views from the deployable actor-axis method. The 0.700 to 0.400 number is a score-cutoff oracle result, not a method result; under raw full-rollout evaluation the same world-frame oracle weakens to 0.700 to 0.600. The method claim must therefore use the stricter same-pass raw rerun in Table 7. In that rerun the oracle proxy hits every frame (1990/1990), and rear-flow risk is active on 312/1990 frames, yet actor-axis selection does not change raw collision rate: 0.700 to 0.700, with zero paired improvements and zero paired regressions. It also leaves offroad unchanged at 0.100, worsens wrong-lane from 0.300 to 0.400, and increases mean distance-to-ground-truth from 1.31 m to 8.55 m. The one robust positive signal is progress, which increases from 0.216 to 0.471 (bootstrap 95% CI [0.083, 0.444] for the paired mean delta). This is useful mechanism evidence, but it is not a deployable actor-aware method.

Table 6: AlpaSim external diagnostic on 10 matched WOD-E2E clips. Rates are fractions of episodes; progress and distances are means. Each intervention repairs a different transfer axis and leaves or worsens another, exposing non-co-monotone simulator-transfer failure rather than a single scalar robustness gap.

Policy	Collision	Offroad	Wrong lane	Progress	Dist. (m)	Dist.-GT
Token-DAgger (2-step)	0.600	0.900	0.700	0.034	61.98	13.86
Token-DAgger (clamped lateral)	0.700	0.500	0.200	0.384	59.85	5.21
Token-DAgger (axis-constrained clamped)	0.700	0.200	0.200	0.358	54.80	2.25
Token-DAgger + hard veto	0.800	0.200	0.700	0.837	166.06	22.71
Token-DAgger (source decay)	0.600	0.900	0.400	0.175	62.92	27.16

Table 7: Matched rear-risk actor-axis rerun on the same 10 AlpaSim clips. Rates are raw full-rollout summaries. Δ collision is the paired better/worse count against the same-pass axis-constrained baseline. The actor-axis proxy has full oracle-proxy timestamp coverage (1990/1990 frames) but remains an inconclusive external method.

Policy	Collision	Offroad	Wrong lane	Progress	Dist. (m)	Dist.-GT	Δ col.
Axis-constrained clamped (matched)	0.700	0.100	0.300	0.216	33.34	1.31	–
Rear-risk actor-axis oracle proxy	0.700	0.100	0.400	0.471	90.91	8.55	0/0

Source-decayed DAgger repairs yet another axis: wrong-lane rate improves from 0.700 to 0.400 and progress improves from 0.034 to 0.175, but collision and offroad rates remain unchanged and distance-to-ground-truth worsens to 27.16 m. Together, these interventions instantiate Proposition 3: under the AlpaSim proxy-state interface, geometry, safety, progress, and lane adherence are not co-monotone. External transfer failure must be diagnosed per axis; a single aggregate pass rate would hide the mechanism.

4.4 Controlled Proxy-State Perturbation

The AlpaSim diagnostic is external and sensor-mediated. To isolate whether the same phenomenon exists without an adapter confound, we run a controlled proxy-transfer experiment inside the internal simulator: physics, scenario dynamics, and token library are held fixed, while only the planner-visible proxy state is perturbed. The safety stack also consumes this perturbed proxy stream, while dynamics and collision evaluation remain true-state. We evaluate lane-scale error, heading bias, actor latency, route offset, and feature noise. The result is a 288-case matched diagnostic (1440 agent rollouts) over adversarial and hidden suites, not a population-level simulator benchmark.

Two cases are especially diagnostic. Under a three-step actor-latency perturbation, raw two-step DAgger falls to 0.604 pass / 0.396 collision / 0.875 lane-violation rate, while the hard-veto hybrid recovers to 0.854 / 0.146 / 0.208 and Spotlight reaches 0.583 / 0.417 / 0.250. This is the clean positive case for grounded suppression: the learned prior and geometric veto together outperform either raw DAgger or Spotlight under delayed actor updates. Route-offset corruption shows the opposite pattern. The same hybrid reduces lane violations from 0.229 to 0.062, but pass rate falls from 0.938 to 0.896 and collision rises from 0.062 to 0.104. Lateral clamping under lane-scale error behaves similarly: lane violations drop from 0.208 to 0.021, while pass rate drops from 0.938 to 0.875 and collision rises from 0.042 to 0.125.

This control is intentionally limited: it is *not* the full simulator oracle. It selects from the same clamped discrete maneuver library as the learned policies, but scores those candidates with privileged short-horizon future-actor forecasts. Its role is to separate token-library insufficiency from token-ordering failure. When oracle-token ordering repairs a perturbation that raw or hybrid selection does not, the candidate set was sufficient and the failure lay in selection under proxy error. The route-offset and lane-scale results provide controlled instances of Proposition 3: an intervention can improve lane adherence while worsening pass or collision, so proxy transfer must be evaluated per axis.

Table 8: Metric provenance for the three actor-aware AlpaSim numbers. The rows answer different questions and must not be read as repeated measurements of one intervention. Only the same-pass raw rear-risk actor-axis row is a deployable-method result.

Probe	Metric view	Collision	Allowed claim
World-frame oracle	Score cutoff	0.700 → 0.400	Actor visibility affects cutoff-era collision accounting; diagnostic only.
World-frame oracle	Raw full rollout	0.700 → 0.600	The oracle effect weakens to one paired raw improvement; inconclusive at $n = 10$.
Rear-risk actor-axis method	Same-pass raw	0.700 → 0.700	Deployable actor-axis proxy is active but not collision-positive.

Table 9: Scene-3 failure vignette from the matched rear-risk actor-axis rerun. Actor-aware signals increase progress and route tracking, but introduce wrong-lane behavior.

Policy	Collision	Offroad	Wrong lane	Progress	Dist.-GT
Axis-constrained clamped	0	0	0	0.185	0.79
Rear-risk actor-axis oracle proxy	0	0	1	0.754	0.22

4.5 WOD-E2E Grounding Placement

The AlpaSim and proxy experiments identify the failing interface: action selection under reconstructed geometry. We therefore test whether frozen latent grounding helps when it is attached to the same interface on the WOD-E2E candidate-selection track. The leverage in this ablation is that all rows evaluate the same 479 validation frames and candidate pool; only the grounding signal and selector head change.

Table 11 provides the strongest connection to recent latent VLA planning work. It does not say that larger latent priors solve transfer by themselves: InternVLA-only features underperform scalar geometry, and linear Cosmos–InternVLA fusion does not close the oracle gap. The positive result appears only when the latent prior is made decision-facing: a nonlinear Cosmos selector reaches 7.845 RFS, and the latent world-prior plus direct selector reaches 7.848 RFS with the lowest tracked regret. This is the same placement principle exposed by AlpaSim: grounding needs to act at the bounded candidate-ranking stage, where externally invalid token choices are made.

5 Analysis

5.1 The Multi-Axis Transfer Result

Table 6 is the central external result. It should not be read as a leaderboard where the hard-veto variant is "best" because it has the highest progress, or where clamping is "best" because it has the lowest distance-to-ground-truth. Those readings collapse a multi-axis transfer problem into a single scalar. The interventions separate the axes:

- **Geometry axis.** Lateral clamping reduces offroad and wrong-lane rates and improves distance-to-ground-truth, showing that part of the transfer error is action magnitude mismatch under reconstructed road geometry. Axis-constrained clamping further lowers offroad and distance-to-ground-truth while leaving collision worse than raw DAgger, separating route adherence from safety.
- **Progress axis.** Hard geometric vetoes greatly increase progress and distance travelled, showing that the DAgger argmax often chooses externally non-progressive tokens.
- **Safety axis.** Neither clamping nor hard vetoes reduce collision rate in the 10-clip matrix; both increase it. Geometry repair and progress repair are therefore not equivalent to safety repair.

Table 10: Controlled internal proxy-state perturbation. Each entry is pass / collision / lane-violation rate. The hybrid is helpful under delayed actors, but route and lane proxy errors expose non-co-monotone tradeoffs.

Perturbation	Raw DAgger	Clamped	Hard veto hybrid	Oracle token	Spotlight
Clean	0.917 / 0.083 / 0.312	0.938 / 0.062 / 0.167	0.958 / 0.042 / 0.167	0.938 / 0.062 / 0.167	0.938 / 0.062 / 0.167
Heading bias	0.896 / 0.104 / 0.562	0.938 / 0.062 / 0.396	0.938 / 0.062 / 0.375	0.896 / 0.104 / 0.292	0.938 / 0.062 / 0.312
Actor latency	0.604 / 0.396 / 0.875	0.604 / 0.396 / 0.521	0.854 / 0.146 / 0.208	0.833 / 0.167 / 0.271	0.583 / 0.417 / 0.250
Route offset	0.938 / 0.062 / 0.229	0.875 / 0.125 / 0.021	0.896 / 0.104 / 0.062	0.938 / 0.042 / 0.125	0.917 / 0.083 / 0.083
Lane-scale error	0.938 / 0.042 / 0.208	0.875 / 0.125 / 0.021	0.958 / 0.042 / 0.083	0.938 / 0.062 / 0.062	0.958 / 0.042 / 0.104
Feature noise	0.917 / 0.083 / 0.917	0.938 / 0.062 / 0.146	0.917 / 0.083 / 0.125	0.896 / 0.104 / 0.104	0.938 / 0.062 / 0.167

Table 11: WOD-E2E grounding-placement ablation on 479 validation frames. RFS is higher better; regret is oracle RFS minus selected RFS. Frozen visual/world priors are not automatically useful: generic fusion remains weak, while selector-aligned nonlinear heads reduce regret.

Ablation	RFS	Oracle	Regret	Match	Grounding signal	World prior	Direct selector
Scalar / geometry only	7.695	9.068	1.373	0.403	none	no	no
InternVLA only	7.672	9.225	1.554	0.392	InternVLA	no	no
Cosmos + InternVLA linear fusion	7.728	9.208	1.480	0.403	Cosmos + InternVLA	no	no
Cosmos 64d nonlinear head	7.845	9.264	1.419	0.390	Cosmos + nonlinear head	no	yes
Cosmos + latent world prior	7.816	9.254	1.438	0.392	Cosmos + latent world prior	yes	no
Cosmos + latent world prior + direct selector	**7.848**	9.254	**1.407**	0.399	Cosmos + latent world prior	yes	yes

- **Actor-completeness axis.** World-frame actor injection is a diagnostic factor, not a solved transfer method: the latest same-pass raw actor-axis rerun has full oracle-proxy timestamp coverage but leaves collision unchanged.
- **Actor-axis method axis.** The explicit rear-risk actor-axis proxy increases progress and distance travelled, but it leaves raw collision/offroad unchanged and worsens wrong-lane. Scaling this variant as a positive method would overstate the evidence.
- **Lane-adherence axis.** Source decay improves wrong-lane rate without changing collision or offroad rate, showing that aggregation calibration affects a different boundary than the geometric veto.

This decomposition is the paper's external-transfer claim. The learned policy is not simply "bad" outside its simulator, and the rule-based scorer is not simply "better." Instead, each correction reveals one latent incompatibility between the internal training geometry and the external proxy-state interface. A robust transfer evaluation must expose these axes separately; otherwise an intervention can appear to improve by moving along one metric while silently degrading another.

5.2 What the Strong Baselines Change

The BC comparison in Table 3 isolates what the token vocabulary does *not* provide. Both Token-BC and Continuous-BC operate on the same 10 geometric features as Spotlight Reflex. Token-BC even uses the identical 9-token vocabulary. Yet both BC agents produce PVs on 3.8–4.4% of Gauntlet scenarios while Spotlight has none observed. Table 4 then shows that recurrent memory alone is not enough: Token-RNN-BC improves PV rate relative to Token-BC, but remains far below Spotlight and two-step DAgger on pass rate.

The failure is mechanistic: on steps where *obstacle_pressure* ≥ 0.35, Spotlight Reflex enforces a hard reference rule—EVASIVE$_{\text{escape}}$ is scored 94 while NUDGE$_{\text{escape}}$ is not offered—committing to full lateral displacement before force cancellation can occur. Token-BC, trained on expert transitions in which MAINTAIN appears 91% of the time, assigns insufficient probability mass to EVASIVE in high-pressure states: even with inverse-frequency class weights, the learned boundary is an approximation that admits errors. Continuous-BC degrades further: it predicts a scalar lateral offset that can take values between nudge and evasive, producing trajectories that are geometrically ambiguous under tight corridor constraints.

Table 12: Three-step DAgger counter-ablations on the same held-out LHS sweep. Removing inverse-frequency token weighting improves offline validation accuracy but breaks closed-loop recovery. Source-decaying later DAgger pools partially restores the frontier without surpassing the two-step policy.

Policy	Overall	Gauntlet	Adversarial	Hidden
Token-DAgger (2-step)	**94.54 / 0.19**	**96.25 / 0.28**	**90.83 / 0.00**	**91.67 / 0.00**
Token-DAgger (3-step)	92.50 / 0.46	94.31 / 0.28	87.92 / 0.42	90.83 / 1.67
Token-DAgger (3-step, source decay)	93.06 / 0.28	94.86 / 0.42	89.17 / 0.00	90.00 / 0.00
Token-DAgger (3-step, no inv. freq.)	89.35 / 2.50	92.08 / 2.08	82.50 / 3.75	86.67 / 2.50
Spotlight Reflex	94.44 / 0.65	96.67 / 0.28	90.00 / 0.83	90.00 / 2.50

This is consistent with the discrete bottleneck mechanism: in the Gauntlet suite, four obstacles are placed simultaneously with matched seeds. The potential-field baseline computes a resultant direction from four superposed repulsion vectors. When obstacles are placed symmetrically or in opposition, the resultant can cancel—a known failure mode of potential-field planners [11]. Token-BC inherits a softer version of the same failure: it *learns* an approximation of when to evade, rather than *computing* it from clearance geometry. Spotlight Reflex's explicit escape-side computation prevents this: $\textit{escape} = \arg\max(\textit{left\_clr}, \textit{right\_clr})$ is always defined and always selects the geometrically correct side.

Two-step DAgger changes the conclusion. Once the learned token policy is trained on states visited by its own first correction, it recovers most of the escape boundary: on the held-out LHS sweep it matches Spotlight overall and improves PV rate. This means the non-interpolation theorem should not be read as saying a neural policy cannot learn the rule. It says the tokenized action space removes unsafe interpolation paths; the empirical question is whether the selector reliably chooses the right token. Plain BC does not. Two-step DAgger mostly does. Three-step DAgger regresses, suggesting that aggregation quality and weighting become part of the safety frontier.

5.3 Deconstructing the Aggregation Collapse

The three-step regression admits at least three plausible explanations: a takeover by late recovery states, a class-frequency optimization bias, or a true calibration failure inside the lightweight token classifier. We tested all three directly.

First, the takeover hypothesis is not supported by the data distribution. The expert BC pool is 91.10% MAINTAIN; DAgger-2 and DAgger-3 remain 92.14% and 91.97% MAINTAIN respectively. The token Jensen–Shannon divergence is only 0.00056 (expert vs. DAgger-2) and 0.00115 (expert vs. DAgger-3). Late DAgger pools therefore do not replace the nominal manifold; they remain close in first-order token statistics.

Second, class-frequency weighting is not the trivial culprit. Removing inverse-frequency token weighting raises offline validation accuracy to 96.3%, but causes a severe closed-loop collapse: three-step DAgger without class weighting drops to 89.35% pass and 2.50% PV overall, with only 82.50% pass and 3.75% PV on Adversarial. This is an offline/online decoupling effect: the unweighted model fits the dominant nominal token mass more accurately while under-serving the rare recovery maneuvers needed in closed loop.

Third, source-dependent weighting partially repairs the failure. We assign loss weights $\alpha = \{1.0, 1.0, 0.5, 0.25\}$ to the expert, first DAgger pool, second DAgger pool, and third DAgger pool respectively, preserving inverse-frequency token weights. The result is a large recovery over the naive three-step model: overall pass improves from 92.50% / 0.46% PV to 93.06% / 0.28% PV, Adversarial recovers from 87.92% / 0.42% PV to 89.17% / 0.00% PV, and Hidden recovers from 90.83% / 1.67% PV to 90.00% / 0.00% PV. Yet this still does not exceed the two-step frontier of 94.54% / 0.19% PV.

These ablations rule out the simplest stories. The three-step failure is not caused by a gross takeover of rare recovery tokens, and it is not fixed by removing class weighting. The evidence instead points to

Table 13: Nested token-count ablation over Gauntlet, Adversarial, and Compositional suites. Each row uses seeds 1–80; PV uses the same failed-rollout $C > 0.90$ criterion as Table 2. The full 9-token set removes observed PVs across the three suites, while reduced sets leave at least one suite with nonzero PVs.

Token set	Gauntlet		Adversarial		Compositional	
	Pass	PV	Pass	PV	Pass	PV
4-token	89.8%	0.6%	90.6%	0.6%	94.2%	0.2%
5-token	94.6%	0.0%	87.5%	0.6%	92.2%	0.2%
7-token	94.0%	0.0%	88.1%	0.0%	91.8%	0.2%
9-token	94.2%	0.0%	91.9%	0.0%	92.2%	0.0%

Table 14: Interaction-aware trajectory-scorer ablation on the same held-out LHS sweep. Entries are pass/PV percentages. Scorers evaluate candidate futures directly, but still trail two-step DAgger on targeted stress suites.

Policy	Gauntlet	Adversarial	Hidden
Interaction Scorer (learner states)	86.81 / 1.94	85.00 / 1.67	89.17 / 1.67
Interaction Scorer (merged states)	88.75 / 1.67	82.92 / 2.50	90.00 / 2.50
Token-DAgger-BC (2-step)	96.25 / 0.28	**90.83 / 0.00**	**91.67 / 0.00**
Spotlight Reflex	96.67 / 0.28	90.00 / 0.83	90.00 / 2.50

a *late-pool aggregation pressure*: once overlapping nominal and recovery pools are mixed uniformly, later recovery states flatten the classifier's boundary. Down-weighting those late pools mitigates the effect, but only partially, consistent with a calibration problem under limited model capacity rather than with a pure data-distribution shift.

Token-count coverage bound. Removing a response class creates a maneuver gap: STOP handles hard blockage within braking distance; CRAWL/MAINTAIN handle low/high-speed forward progress; SLOW_YIELD handles yield-without-stopping; $\text{NUDGE}_{\text{L/R}}$ handles corridor bias without full lane departure; $\text{EVASIVE}_{\text{L/R}}$ handles head-on and cut-in scenarios requiring larger lateral displacement; LANE_RECOVER handles post-evasion re-centering. Table 13 reports a nested token-count ablation. It supports the claim that the full token set closes observed PV gaps in this simulator, but it does not prove global optimality of nine tokens.

5.4 Trajectory-Scorer Ablation

A natural alternative to hard-coded scoring is to learn a utility function over the same candidate trajectories. We tested trajectory-informed scorers that evaluate each token's short-horizon ego rollout, augmented with kinematic invariants (curvature, heading deltas, path length, lateral-acceleration and jerk proxies) and explicit interaction features (minimum dynamic clearance, time to closest approach, signed longitudinal/lateral miss distances, relative speed, and crossing relation). These models are trained contrastively: the Spotlight-selected candidate is the positive and the other token candidates are negatives.

Table 14 weakens a simpler explanation: in this setup, the remaining gap is not closed by replacing the flat token head with a candidate utility scorer over ego trajectories plus lightweight interaction features. The scorer sees the candidate futures, but it is still an offline selector. In closed loop it trails the two-step DAgger policy on Gauntlet and Adversarial and does not preserve the zero-PV hidden result. This points to closed-loop state coverage as the dominant ingredient in the current simulator, rather than a need for the tested per-candidate scalar scorers.

5.5 Failure Mode Taxonomy

The 28 Gauntlet non-passes by Spotlight Reflex (Table 2; 480 scenarios, 94.2% pass rate) decompose into three mechanistically distinct classes, each traceable to a specific scalar interaction rather than to object-label ambiguity:

Over-stopping under cumulative pressure (46% of failures). Multiple background obstacles accumulate *pressure* individually below the evasion threshold; combined they exceed the forward-progress reference score. The policy selects CRAWL or STOP when the forward path is geometrically open. Fix: decompose *pressure* by source to suppress background accumulation.

Slow re-entry after evasion (30% of failures). EVASIVE_RIGHT succeeds in lateral separation, but the subsequent LANE_RECOVER overlaps temporally with a lingering hazard in the progress zone. The progress gate penalizes re-entry, causing a stall. Fix: widen the progress gate temporal window post-evasion.

Synchronized hazard overload (24% of failures). Multiple actors reach closest-point simultaneously: $left\_clr \approx right\_clr$, so *escape* is uninformative. Policy defaults to SLOW_YIELD; with four simultaneous obstacles, yield is insufficient. Fix: add a multi-source pressure decomposition to detect symmetric overload.

In all three cases the failure is traceable to a scalar computation, not to an object label—consistent with the geometry-invariance property. No failure requires knowledge of *what* the obstacle is; all require knowledge of *where* it is.

6 Limitations and Future Work

2D abstract simulation. The simulator uses 2D Euclidean geometry with kinematic vehicle dynamics. The AlpaSim diagnostic (Sec. 4.3) shows that internal simulator success is not a transfer guarantee: the learned selector's calibration can fail when the same token vocabulary is driven by reconstructed external geometry. The 10-clip AlpaSim matrix is mechanistic evidence from matched external clips and selection logs, not a population estimate. A full physics simulation (e.g., CARLA, nuPlan with dynamics) or real-vehicle evaluation is needed before making broad vehicle-dynamics claims.

Token-count optimality is not established. Table 13 is a nested removal study, not a search over all token libraries. The stronger claim that nine is near-minimal requires random token libraries, learned token sets, parameter perturbations of offsets and speed scales, and larger token counts (e.g., 12-token variants) to test whether extra classes induce token-flip instability.

Stress-test operating envelope (empirical). To quantify the *operating boundary* of geometric reflex vs. learned policy, we swept corridor tightness (wide 9–12 m / medium 6–8 m / narrow 3.6–5 m / extreme 2.2–3.6 m) × obstacle-actor pressure (1.0× / 1.5× / 2.0× of the Gauntlet baseline), 720 rollouts per cell per agent (25,920 total). Table 15 summarises the PV rates.

Table 15: Phase diagram: worst-case Proximity Violation (PV) rate per corridor level (max over 3 pressure settings, 720 rollouts each). Spotlight Reflex ≤0.4% PV across all 12 conditions; BC peaks at 6.5% in narrow/extreme corridors. Full 12-cell results in the artifact.

Corridor	ContBC	TokBC	**Spotlight**	ΔTok–SR
Wide (9–12 m)	4.3%	3.9%	**0.4%**	+3.9 pp
Medium (6–8 m)	4.2%	3.6%	**0.1%**	+3.5 pp
Narrow (3.6–5 m)	5.6%	6.5%	**0.3%**	+6.4 pp
Extreme (2.2–3.6 m)	4.4%	6.4%	**0.3%**	+6.4 pp

Spotlight Reflex holds $\leq$0.4% PV rate in all 12 cells. BC agents degrade as corridors narrow: the Token-BC delta grows from +2.2 pp (wide, 2.0×) to +6.4 pp (narrow/extreme, 1.0×/1.5×). BC agents were trained on compositional and adversarial suites with default wide-lane topologies (6–9 m half-widths); the narrow and extreme cells push *left_clearance* and *right_clearance* outside the training distribution, and the learned boundary degrades. Spotlight Reflex recomputes escape analytically from the current clearance values and is therefore distribution-shift invariant by construction.

This phase diagram should be read as an operating-envelope visualization, not as the primary environment proof. The stronger evidence is the held-out Latin-hypercube sweep in Sec. 4.2, which independently samples generator parameters and includes recurrent BC plus DAgger-style on-policy correction.

Notably, Token-BC *exceeds* Continuous-BC at extreme OOD (4.4% vs 6.1–6.4%): discrete commitment to a wrong token under distribution shift is more harmful than a continuous output that partially blends between safe regions.

Temporal scalar history and joint interaction. The strongest learned model in this paper is still a token classifier over 10 scalar features. Token-RNN-BC adds scalar history but remains weak, and interaction-aware trajectory scorers do not close the stress-suite gap. A richer joint future encoder over ego and actor rollouts may be needed to distinguish coordinated interaction from static clearance geometry.

DAgger aggregation is not solved. Two-step DAgger is the strongest baseline, but three-step DAgger regresses. This suggests that blindly adding recovery-state data can distort the training distribution. Future work should study aggregation schedules, loss weighting by rollout source, validation splits by closed-loop profile, and safety-constrained selection among DAgger checkpoints.

External calibration is not solved. The AlpaSim hard-veto experiment shows that a learned selector can be internally strong and externally invalid: 1987/1990 DAgger argmax decisions are rejected by the geometric suppression layer. This is not a threshold-tuning problem; it indicates that proxy-state reconstruction changes the decision geometry. The 10-clip matrix further shows that interventions are not co-monotone across collision, offroad, wrong-lane, progress, and tracking error. The actor-aware external result is also limited: even with 1990/1990 oracle-proxy timestamp hits and rear-risk activations on 312/1990 frames, the latest matched raw actor-axis rerun leaves collision unchanged and worsens wrong-lane while improving progress. Future work should validate token policies under multiple independent simulators, log per-frame disagreement with geometric constraints, separate rear/at-fault collisions, and report per-axis failure metrics rather than tuning one aggregate score.

7 Conclusion

We have argued that discrete maneuver tokens are useful, but not sufficient. Internally, tokens remove unsafe interpolation paths, geometric rules provide an interpretable expert, and two-step DAgger can learn most of that expert's closed-loop boundary on a held-out Latin-hypercube generator sweep. That internal result alone would support a conventional methods story: on-policy correction repairs plain BC and nearly matches the geometric policy.

The AlpaSim diagnostic changes the conclusion. When the learned selector is driven by a front-camera proxy adapter, its token calibration no longer matches the external geometry, and the resulting transfer error is multi-axis. Lateral clamping repairs offroad and wrong-lane behavior but increases collision rate. Hard geometric suppression repairs progress and offroad behavior but vetoes 1987/1990 learned argmax decisions, raises collision rate, and leaves wrong-lane rate unchanged. Source decay improves wrong-lane rate but does not repair offroad or collision. The actor-aware proxy then shows the same pattern: full oracle-proxy coverage and rear-risk signals increase progress, but do not reduce raw collision and worsen wrong-lane. The useful finding is therefore not that DAgger plus a veto layer or actor-axis proxy is a winning method. It is that offline accuracy and internal closed-loop performance can be structurally misleading proxies for external robustness when the state reconstruction interface changes the metric ordering of the decision problem.

The practical implication is that token policies should be audited at the decision level and per failure axis, not only by aggregate rollout metrics. A learned policy that passes an internal simulator may still be selecting tokens that an external geometric scorer rejects almost every frame. For safety-critical imitation learning, per-frame constraint disagreement, external proxy-state validation, and separate geometry/safety/progress/lane metrics should be treated as first-class diagnostics alongside pass rate and offline validation accuracy.

Acknowledgments

Developed independently as a research prototype for the Waymo Open Dataset E2E Challenge. We thank the Waymo team for the AlpaSim evaluation environment.

References

[1] D. Dauner, M. Hallgarten, A. Geiger, and K. Chitta. Parting with misconceptions about learning-based vehicle motion planning. In *Conference on Robot Learning*, 2023.

[2] Y. Hu, J. Yang, L. Chen, K. Li, C. Sima, X. Zhu, S. Chai, S. Du, T. Lin, W. Wang, et al. Planning-oriented autonomous driving. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 17853–17862, 2023.

[3] K. Chitta, A. Prakash, B. Jaeger, Z. Yu, K. Renz, and A. Geiger. TransFuser: Imitation with transformer-based sensor fusion for autonomous driving. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.

[4] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski, T. Ding, D. Driess, A. Dubey, C. Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.

[5] D. A. Pomerleau. ALVINN: An autonomous land vehicle in a neural network. In *Advances in Neural Information Processing Systems*, volume 1, 1989.

[6] S. Ross, G. Gordon, and D. Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *International Conference on Artificial Intelligence and Statistics*, pages 627–635, 2011.

[7] J. Ho and S. Ermon. Generative adversarial imitation learning. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

[8] M. Bansal, A. Krizhevsky, and A. Ogale. ChauffeurNet: Learning to drive by imitating the best and synthesizing the worst. In *Robotics: Science and Systems*, 2019.

[9] H. Caesar, J. Kabzan, K. S. Tan, W. K. Fong, E. Wolff, A. Lang, L. Fletcher, O. Beijbom, and S. Omari. nuPlan: A closed-loop ml-based planning benchmark for autonomous vehicles. In *CVPR Workshop on Autonomous Driving*, 2021.

[10] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.

[11] O. Khatib. Real-time obstacle avoidance for manipulators and mobile robots. *The International Journal of Robotics Research*, 5(1):90–98, 1986.

[12] F. Codevilla, M. Müller, A. López, V. Koltun, and A. Dosovitskiy. End-to-end driving via conditional imitation learning. In *IEEE International Conference on Robotics and Automation*, pages 4693–4700, 2018.

[13] Y. Luo, F. Li, S. Xu, Y. Ji, Z. Zhang, B. Wang, Y. Shen, J. Cui, L. Chen, G. Chen, H. Ye, Z.-X. Yang, and F. Wen. LaST-VLA: Thinking in latent spatio-temporal space for vision-language-action in autonomous driving. *arXiv preprint arXiv:2603.01928*, 2026.

[14] T. Khurana, P. Hu, D. Held, and D. Ramanan. Differentiable raycasting for self-supervised occupancy forecasting. In *European Conference on Computer Vision*, pages 353–369, 2022.

[15] C. Shi, Z. Jiang, Z. Zhao, H. He, and Y. Fu. MotionDiffuser: Controllable multi-agent motion prediction using diffusion. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2024.